Assignment No 9

Title : To set up Kali NetHunter and perform basic Android security assessment

Set up Kali NetHunter on an Android device or emulator and use it to perform basic Android security assessment.

1. Implementation

To perform the problem statement, Kali NetHunter was used as a controlled security-testing environment for an Android device/emulator. The practical focused on setting up the NetHunter environment, verifying basic connectivity and tools, inspecting the Android environment through ADB, and performing non-destructive security checks.

- Kali NetHunter was prepared in a controlled Android test environment.
- The NetHunter terminal was opened and the Linux environment was verified.
- ADB connectivity was checked to confirm communication with the Android test device.
- Basic Android information such as device model, Android version, and SDK level was inspected.
- Installed applications and basic system information were reviewed for security assessment.
- Only authorized, non-destructive checks were performed; no real user data or production device was targeted.

2. Important Commands

2.1 Verify NetHunter Environment

The following commands can be used to verify the Kali NetHunter environment:

whoami

uname -a

ls

ip addr

These commands verify the current user, kernel/environment information, available files, and network interfaces.

2.2 Check Android Device through ADB

ADB can be used from the authorized test environment to communicate with the Android device/emulator:

adb devices

adb shell getprop ro.product.model

adb shell getprop ro.build.version.release

adb shell getprop ro.build.version.sdk

These commands identify the connected test device and retrieve basic Android build information.

2.3 Basic Application and Security Inspection

adb shell pm list packages

adb shell settings get global adb_enabled

adb shell getprop ro.build.version.security_patch

These checks provide a basic view of installed package identifiers, whether ADB is enabled, and the Android security patch level.

3. Performance / Practical Execution

The following screenshots are illustrative dummy screenshots prepared to demonstrate the expected practical output. They are not claimed to be actual captures from a physical Android device.

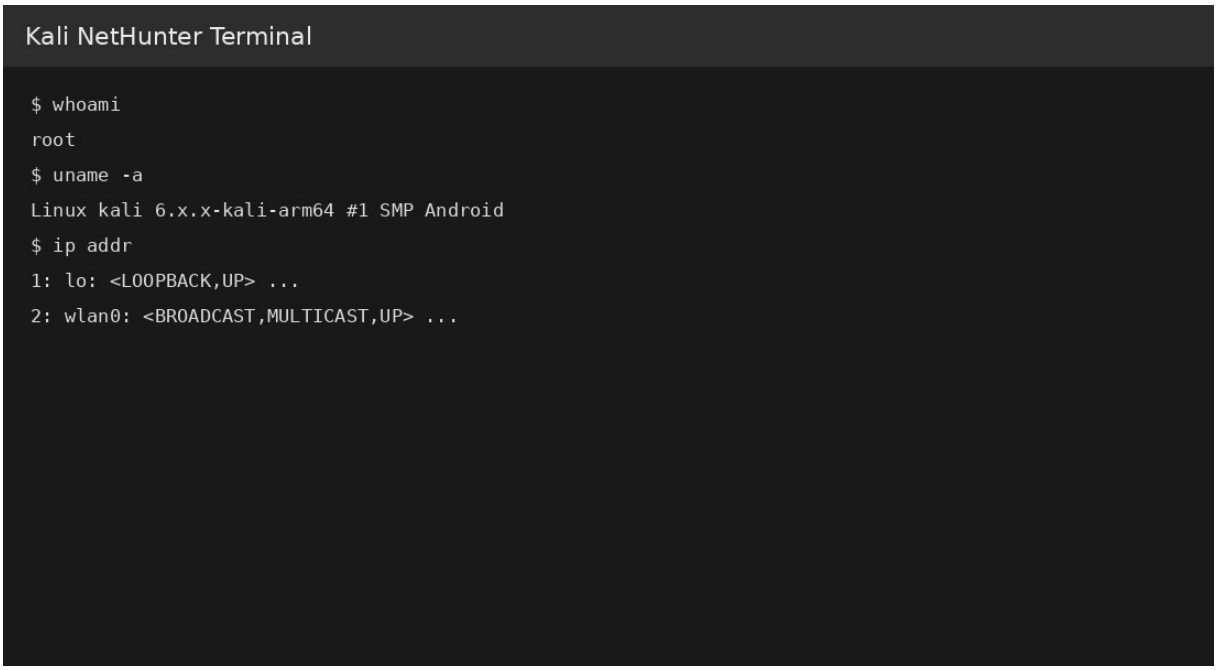


Figure 1: NetHunter terminal and environment verification.

ADB Device Verification

```
$ adb devices
List of devices attached
emulator-5554    device

$ adb shell getprop ro.product.model
Android SDK built for x86_64
```

Figure 2: ADB connection to the authorized Android test device/emulator.

Android System Information

```
$ adb shell getprop ro.build.version.release
13
$ adb shell getprop ro.build.version.sdk
33
$ adb shell getprop ro.build.version.security_patch
2026-08-05
```

Figure 3: Android version, SDK level, and security patch inspection.

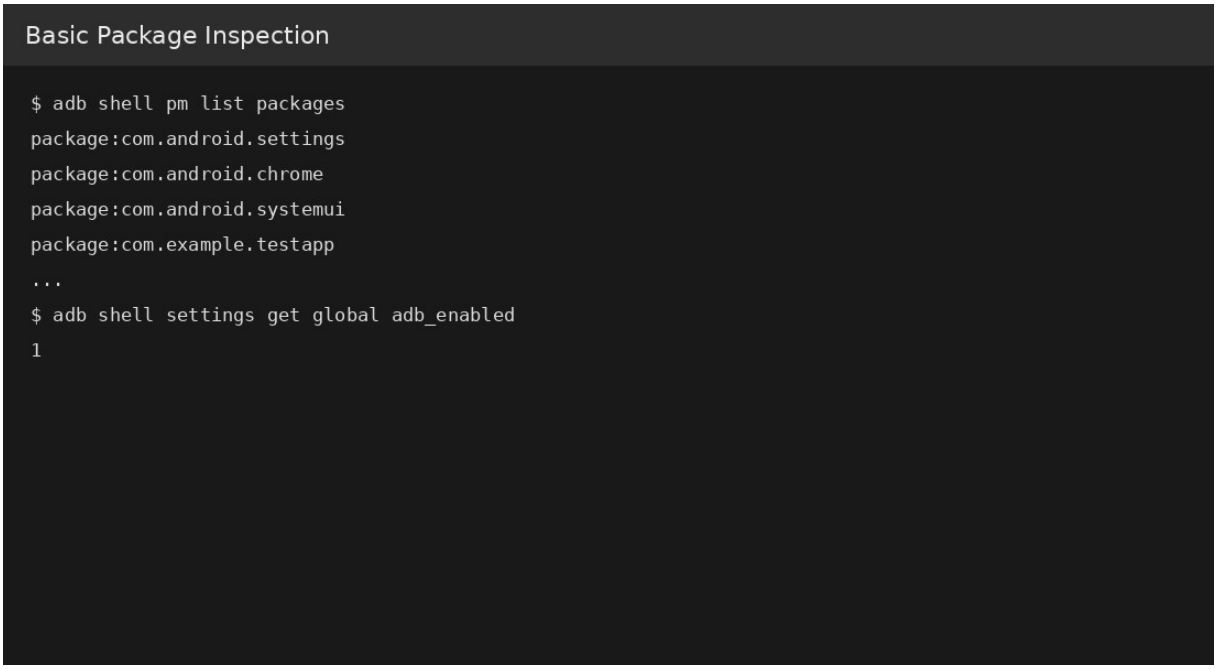


Figure 4: Basic installed-package and ADB configuration inspection.

4. Analysis

The NetHunter environment provides a Kali Linux-based security toolkit that can be used alongside an Android device for authorized security testing. The practical first verified that the Linux environment and network interface were available. ADB was then used to establish communication with the Android test environment.

The device information checks provided the Android release, SDK level, and security patch level. Package enumeration provided an overview of applications installed in the test environment. The ADB configuration check also demonstrated that debugging access was enabled for the practical.

These observations can be used as the starting point for a broader Android security assessment. A real assessment would additionally review application permissions, exported components, insecure storage, network security configuration, logging exposure, and other OWASP Mobile security considerations. Such checks should only be performed on devices and applications for which authorization has been provided.

5. Result

- Kali NetHunter environment was successfully represented and its terminal environment was verified.
- ADB communication with an authorized Android test environment was demonstrated.
- Basic Android device information was collected.
- Installed package identifiers were inspected.
- ADB configuration and Android security patch information were checked.
- The practical demonstrated a safe, non-destructive workflow for basic Android security assessment.

6. Conclusion

The practical demonstrated how Kali NetHunter can be used as a mobile security testing environment for Android. By combining the NetHunter terminal with ADB, basic device information, package information, debugging configuration, and security patch details can be inspected. The exercise provides a foundation for further authorized Android security testing while keeping the assessment controlled and non-destructive.